

Misael Vargas

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EDUCATION

Texas Tech University
BS in Mechanical Engineering
GPA: 3.58
Portfolio: <https://mvargas001.github.io/Misael-Portfolio/>

Lubbock, Tx
Aug 2024 - May 2028

WORK EXPERIENCE

Transland, LLC
Engineering Intern

Wichita Falls, TX
June 2026 – August 2026

- Rebuilt an inaccurate AT-802 wing fuel-volume model, cutting error from ~20–30 gal to ~2 gal by tracing a flat 2D section, validating against hardware and catching errors in the prior CAD models through drawings.
- Delivered 30+ SolidWorks fuel tank configurations (min/median/max fuel levels) via an Excel-driven workflow, with fuel-gauge coordinates for the drawings.
- Reverse-engineered a legacy hopper-cart assembly in SolidWorks, designing original parts and running hand statics; specified materials, welds, and fasteners with safety factors, and owned the manufacturing hand-off.

Tortilleria Lupita
Co-Owner & Operations Manager

Wichita Falls, TX
Oct 2018 – June 2026

- Directed a team of 7 in a high-tempo production environment, managing daily workflow and peak-period logistics to serve 100+ clients per shift while enforcing quality standards.
- Managed supply chain logistics, equipment maintenance, and strict quality assurance protocols for high-volume daily output

PROJECTS & RESEARCH

Decurion: Human – Machine Teaming

Personal Project
June 2026 - present

Multi-drone command system: voice control, MAVLink command stack, and hardware integration

- Designed a Python/MAVLink command stack enabling one operator to direct two autonomous aircraft (waypoint navigation and a ~5m FOLLOW mode), fully validated in ArduPilot SITL.
- Built a non-blocking, offline voice interface (Faster-Whisper) with a two-stage command parser and multi-drone routing, utilizing multi-threading for seamless real-time execution.
- Integrated a 6S Ardupilot quadcopter (STM32, ELRS, LoRa) and executed root-cause analysis on ESC-to-FC power failures, authoring a standardized pre-power checklist that eliminated hardware loss.

Undergrad Research Assistant
Smart Human-centric Automation Resilience Lab

Texas Tech University
September 2026 - present

- Selected to assist a graduate researcher in developing a custom driving simulation designed to compare human and autonomous vehicle (AV) responses in high-risk traffic scenarios.
- Currently training in Python and simulation architecture to prepare for the development of dynamic variables, such as traffic congestion and unpredictable driver behaviors.
- Reviewing project requirements and shadowing graduate researchers to support the upcoming design of the experimental testbed for human-automation interaction trials.

LEADERSHIP & EXPERIENCE

SHPE (Society of Hispanic Professional Engineers) - Texas Tech Chapter
Member

Lubbock, TX
Oct 2025 – Feb 2026

- Coordinated outreach and volunteer programs to facilitate technical growth through workshops and corporate networking for chapter members.

American Society of Mechanical Engineering
Member

Lubbock, TX
Aug 2026 – Present

- Attending technical seminars and professional development events to gain exposure to industry practices and mechanical engineering trends.

TECHNICAL SKILLS

- CAD & Design: SolidWorks (CSWA), GD&T, Manufacturing Hand-off, Stress & Interference analysis
- Programming & Tools: Python, MAVLink/Pymavlink, Ardupilot, SITL, Excel
- Hardware & Prototyping: Electronics Soldering, Multimeter Diagnostics, 3D printing, Root Cause Analysis